Multivariable calculus · Differentiation in two variables

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Partial derivatives, changes along a path, and a circle defined implicitly · compiled 17 October 2026

S1	Lesson: partial derivatives
S2	Lesson: total derivative
S3	Lesson: implicit curves

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A Formula sheet

1 Partial derivatives and directional slopes

E1.1 · Coordinate derivatives

$$f_x = 2xy, \qquad f_y = x^2 + \cos y$$

gives	Coordinate slopes for $f = x^2 y + \sin y$.
applies	Hold the other independent coordinate fixed.
units	Output/input; dimensionless in this example.

S1

E1.2 · Directional slope

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$$

gives	Slope in a selected direction.
applies	f differentiable and $\|\mathbf{u}\| = 1$.
units	Output per unit distance.

S1

E1.3 · Second partials

$$f_{xx} = 2y, \quad f_{xy} = f_{yx} = 2x$$
$$f_{yy} = -\sin y$$

GIVES Variation of coordinate slopes; these are second partial derivatives.

APPLIES For the stated smooth function.

UNITS Output/input squared.

S1

2 Total change and the chain rule

E2.1 · Differential

$$df = f_x\,dx + f_y\,dy$$

GIVES First-order change.

APPLIES The function is differentiable at the base point. A finite increment is approximated by this differential.

UNITS Same as the output.

S2

E2.2 · Chain rule

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}$$

GIVES Change along a parameterised path.

APPLIES Differentiate every changing input. Evaluate partials on the path.

UNITS Output per parameter unit.

S2

3 Implicit curves and valid tangent formulas

E3.1 · Implicit first derivative

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

GIVES Slope of a regular implicit branch.

APPLIES $F = 0$ at the point and $F_y \neq 0$.

UNITS Units of y divided by units of x.

S3

E3.2 · Circle second derivative

$$y'' = -\frac{25}{y^3}$$

GIVES	Second derivative of a branch of $x^2 + y^2 = 25$.
APPLIES	$y \neq 0$; this formula is specific to the radius-five circle.
UNITS	Dimensionless coordinates.

S3

B Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
f_x	Partial derivative in x	Change in f per change in x while y stays fixed. S1	output/x	E1.1
$\mathbf{u}$	Unit direction	Chosen vector of length one. S1	dimensionless	E1.2
df	Differential	Linear part of output change. S2	output	E2.1
t	Path parameter	Independent variable along a path. S2	parameter unit	E2.2
F	Defining function	The relation is the level set F=0. S3	context-dependent	E3.1
x, y	Independent coordinates	The two real input coordinates of f. S1	Dimensionless	E1.1
f	Scalar function	In this lesson, $f(x, y) = x^2 y + \sin y$. S1	Dimensionless	E1.1

Symbol	Name	Meaning	Unit	Appears in
∇f	Gradient	The vector (f_x, f_y); its dot product with a unit vector gives the directional derivative. S1	Output per input	E1.2
$D_{\mathbf{u}} f$	Directional derivative	The rate of change per unit displacement in the unit direction $\mathbf{u}$. S1	Output per input	E1.2
y', y''	Implicit derivatives	First and second derivatives of a local branch $y(x)$, where that branch exists. S3	Output per input; output per input squared	E3.1, E3.2

Remark. The symbols x and y denote independent coordinates except when a specific path or implicit branch is stated.